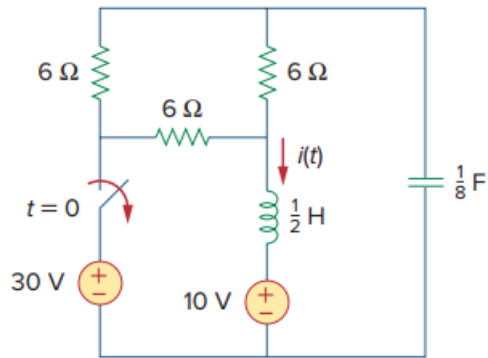


Problem

For the network in Fig. 8.85, solve for $i(t)$ for $t > 0$.

Figure 8.85:

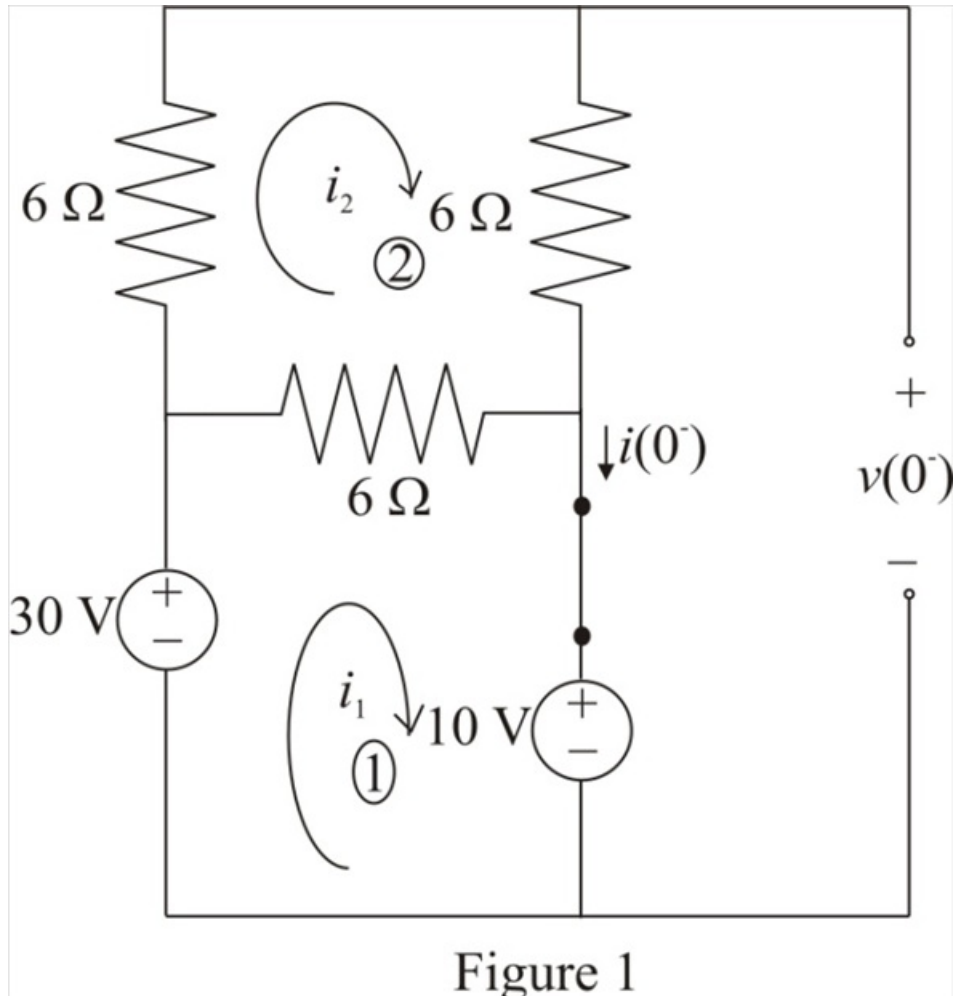


Step-by-step solution

Step 1 of 16

Refer to Figure 8.85 in the text book.

At $t < 0$, the capacitor acts as open circuit and inductor acts as short circuit.



[Comments \(1\)](#)



Anonymous

Wrong. Capacitors act as short circuits and inductors act as open after time t

Step 2 of 16

Apply Kirchhoff's Voltage Law in loop-2.

$$6i_2 + 6i_2 + 6(i_2 - i_1) = 0$$

$$i_2 + i_2 + i_2 - i_1 = 0$$

$$3i_2 - i_1 = 0$$

$$i_1 = 3i_2 \text{ (1)}$$

Therefore, the expression for the current i_1 is $3i_2$.

Step 3 of 16

Apply Kirchhoff's Voltage Law in loop-1.

$$-30 + 6(i_1 - i_2) + 10 = 0$$

$$6i_1 - 6i_2 - 20 = 0$$

Substitute $3i_2$ for i_1 .

$$6(3i_2) - 6i_2 - 20 = 0$$

$$18i_2 - 6i_2 - 20 = 0$$

$$12i_2 - 20 = 0$$

$$12i_2 = 20$$

$$i_2 = \frac{20}{12}$$

$$= 1.6667 \text{ A}$$

Therefore, the value of the current i_2 is 1.6667 A .

Step 4 of 16

Recall equation (1).

$$i_1 = 3i_2$$

Substitute 1.6667 A for i_2 .

$$i_1 = 3(1.6667)$$

$$= 5 \text{ A}$$

Here, i_1 represents the current $i(0^-)$.

$$i(0^-) = 5 \text{ A}$$

Therefore, the value of the current $i(0^-)$ is 5 A .

Step 5 of 16

Calculate the value of the voltage $v(0^-)$.

$$v(0^-) = 6i_2 + 10$$

Substitute 1.6667 A for i_2 .

$$v(0^-) = 6(1.6667) + 10$$

$$= 10 + 10$$

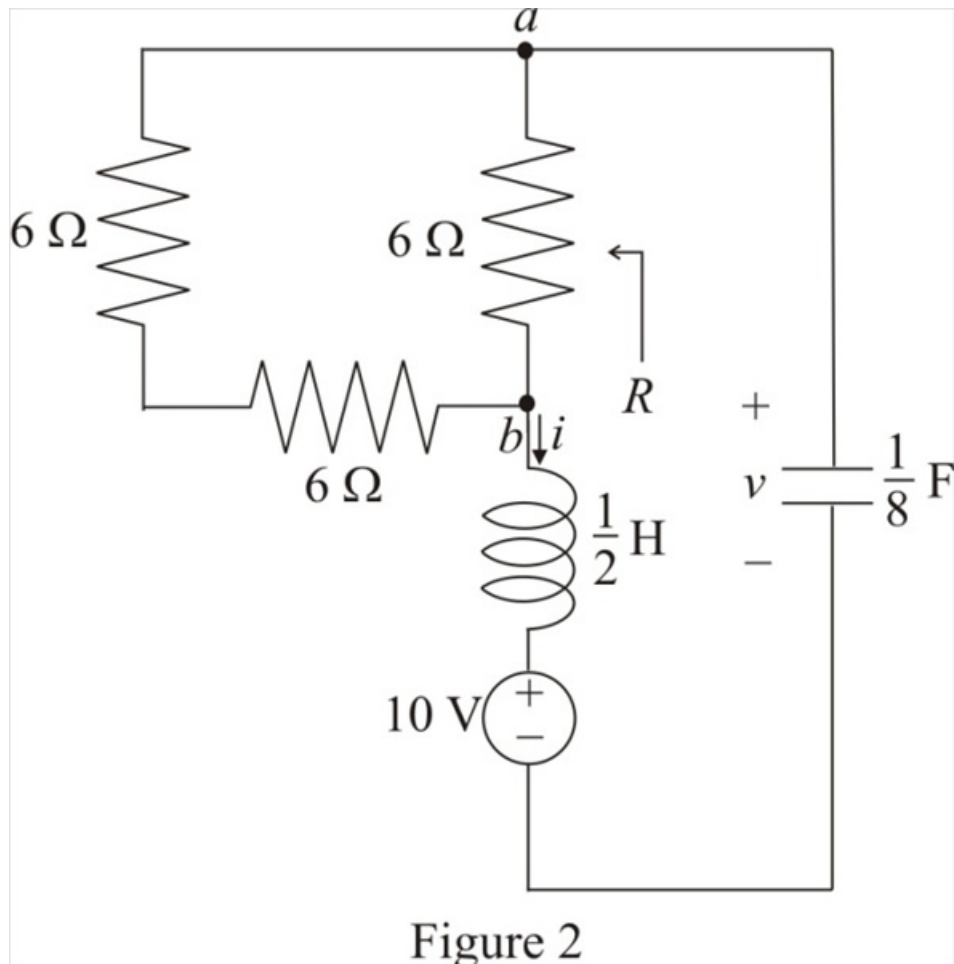
$$= 20 \text{ V}$$

Therefore, the value of the voltage $v(0^-)$ is 20 V .

Step 6 of 16

The switch opens at $t > 0$.

Draw the circuit diagram at $t > 0$.



Step 7 of 16

Calculate the value of the resistance as seen from the terminals a and b .

$$\begin{aligned} R &= (6+6) \parallel 6 \\ &= 12 \parallel 6 \\ &= \frac{(12)(6)}{12+6} \\ &= \frac{72}{18} \end{aligned}$$

$$R = 4 \, \Omega$$

Therefore, the value of the resistance R is $4 \, \Omega$.

Step 8 of 16

Apply Kirchhoff's Voltage Law in the loop.

$$iR + 10 + \frac{1}{2} \frac{di}{dt} - v = 0$$

Substitute 4Ω for R .

$$4i + 10 + \frac{1}{2} \frac{di}{dt} - v = 0$$

$$\frac{1}{2} \frac{di}{dt} = v - 10 - 4i$$

$$\frac{di}{dt} = 2v - 20 - 8i$$

Calculate the value of $\frac{di}{dt}$ for $t = 0$.

$$\frac{di(0)}{dt} = 2v(0) - 20 - 8i(0)$$

Substitute 5 A for $i(0)$ and 20 V for $v(0)$.

$$\begin{aligned} \frac{di(0)}{dt} &= 2(20) - 20 - 8(5) \\ &= 40 - 20 - 40 \\ &= -20 \text{ A/s} \end{aligned}$$

Therefore, the value of $\frac{di(0)}{dt}$ is -20 A/s .

Step 9 of 16

Deactivate the sources from the circuit diagram to estimate the transient response.

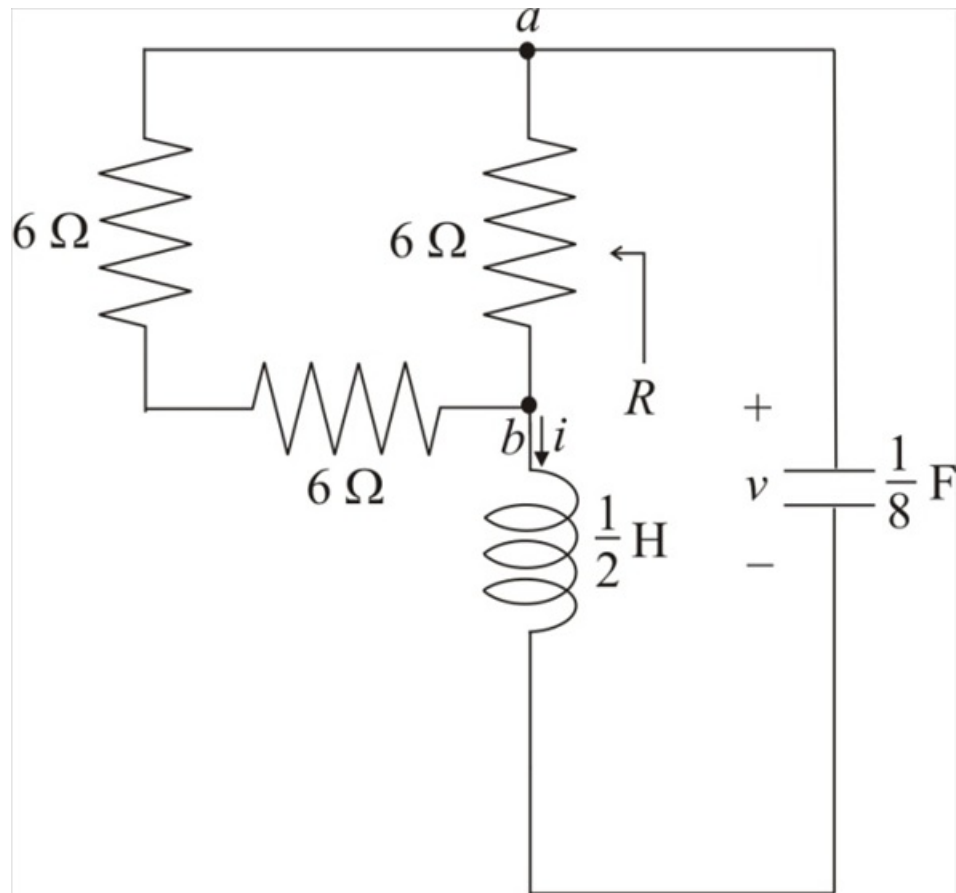


Figure 3

Step 10 of 16

Apply Kirchhoff's Voltage Law in the loop.

$$iR + \frac{1}{2} \frac{di}{dt} + \frac{1}{1/8} \int i \, dt = 0$$

Substitute $4 \, \Omega$ for R .

$$4i + \frac{1}{2} \frac{di}{dt} + 8 \int i \, dt = 0$$

Apply derivative with respect to t .

$$4 \frac{di}{dt} + \frac{1}{2} \frac{d^2i}{dt^2} + 8i = 0$$

$$\frac{d^2i}{dt^2} + 8 \frac{di}{dt} + 16i = 0$$

Consider $\frac{d}{dt}$ to be s .

$$s^2i + 8si + 16i = 0$$

$$(s^2 + 8s + 16)i = 0$$

[Comments \(1\)](#)



Anonymous

shouldn't be negative the voltage drop across the inductor ? if you follow Ohm's law then your KVL is wrong

Step 11 of 16

Consider the characteristic equation of the circuit.

$$s^2 + 8s + 16 = 0$$

$$(s + 4)^2 = 0$$

$$s + 4 = 0$$

$$s = -4$$

The circuit exhibits critically damped response since the roots are negative and equal.

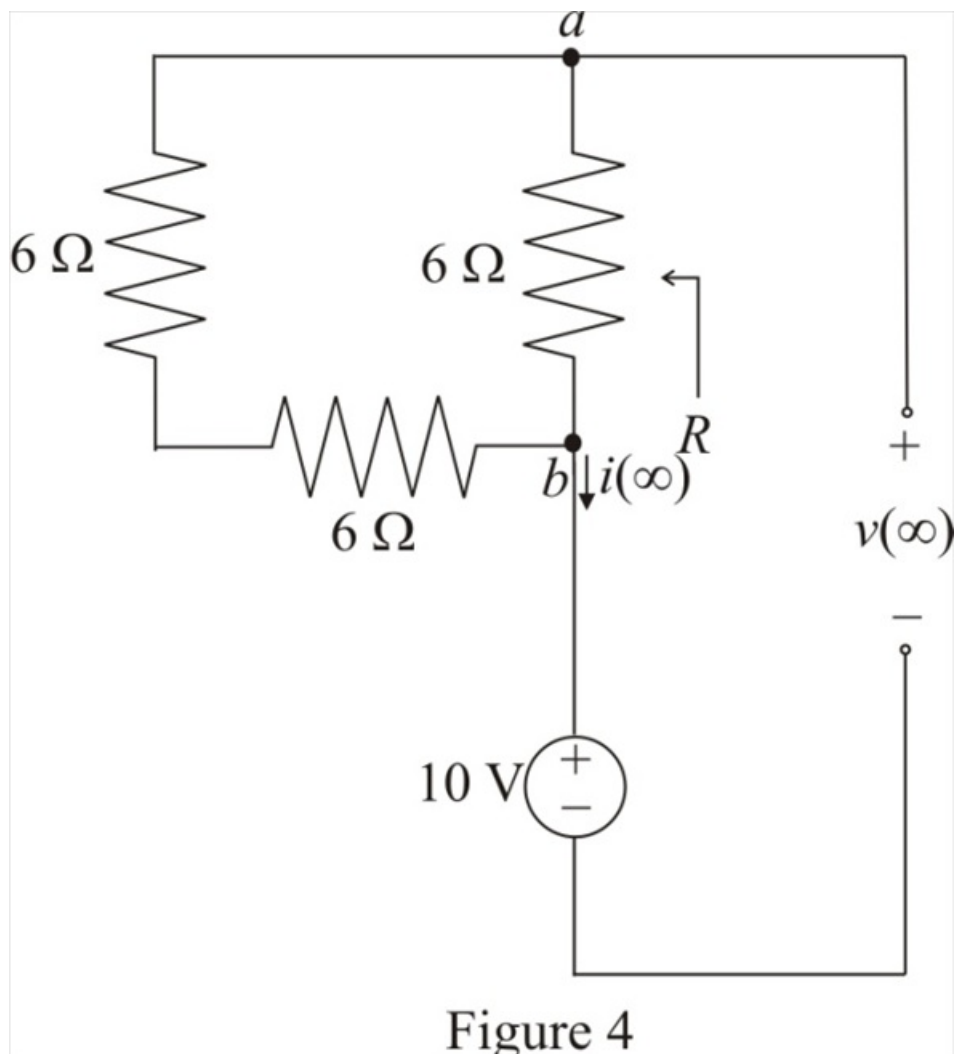
Compare with $s = \alpha$.

$$\alpha = -4$$

Therefore, the value of α is -4

Step 12 of 16

Draw the circuit at $t = \infty$.



The value of the current $i(\infty)$ is 0 A

Step 13 of 16

Consider the general solution for the current for the critically damped response.

$$i(t) = i(\infty) + (A_1 + A_2 t)e^{\alpha t}$$

Substitute 0 A for $i(\infty)$ and -4 for α .

$$i(t) = 0 + (A_1 + A_2 t)e^{-4t}$$

$$i(t) = (A_1 + A_2 t)e^{-4t} \dots\dots (1)$$

Step 14 of 16

Recall equation (1).

$$i(t) = (A_1 + A_2 t) e^{-4t}$$

Substitute 0 for t .

$$\begin{aligned} i(0) &= (A_1 + A_2(0)) e^{-4(0)} \\ &= (A_1 + 0) e^{(0)} \\ &= A_1(1) \\ &= A_1 \end{aligned}$$

Substitute 5 A for $i(0^-)$.

$A_1 = 5$ Therefore, the value of the constant A_1 is 5.

Step 15 of 16

Recall equation (1).

$$i(t) = (A_1 + A_2 t) e^{-4t}$$

Apply derivative with respect to t .

$$\begin{aligned} \frac{di(t)}{dt} &= \frac{d}{dt} \{ (A_1 + A_2 t) e^{-4t} \} \\ &= (A_1 + A_2 t) \frac{d}{dt} e^{-4t} + e^{-4t} \frac{d}{dt} (A_1 + A_2 t) \\ &= (A_1 + A_2 t) (-4) e^{-4t} + e^{-4t} (0 + A_2) \\ &= (A_1 + A_2 t) (-4) e^{-4t} + A_2 e^{-4t} \end{aligned}$$

Substitute 0 for t .

$$\begin{aligned} \frac{di(0)}{dt} &= (A_1 + A_2(0)) (-4) e^{-4(0)} + A_2 e^{-4(0)} \\ &= (A_1 + 0) (-4) e^{(0)} + A_2 e^{(0)} \\ &= (-4A_1)(1) + A_2(1) \\ &= -4A_1 + A_2 \end{aligned}$$

Substitute 5 for A_1 and -20 A/s for $\frac{di(0)}{dt}$.

$$\begin{aligned} -20 &= -4(5) + A_2 \\ -20 &= -20 + A_2 \\ A_2 &= -20 + 20 \\ A_2 &= 0 \end{aligned}$$

Therefore, the value of the constant A_2 is 0.

Step 16 of 16

Recall equation (1).

$$i(t) = (A_1 + A_2 t) e^{-4t}$$

Substitute 5 for A_1 and 0 for A_2 .

$$\begin{aligned} i(t) &= (5 + (0)t) e^{-4t} \\ &= (5 + 0) e^{-4t} \\ &= 5e^{-4t} \text{ A} \end{aligned}$$

Therefore, the value of the current $i(t)$ is $5e^{-4t}$ A.